For the submission of your work:

- Create a folder named **RollNo_Name_DBI202_PaperNo**, e.g. he170145_LongNT_DBI202_01. **Do not** create any subfolder in this folder. All file created will be located in the above folder.

- For each question, you are required to write a database script. Create a file with the name corresponding to the index of the question. For example, **for question 1**, we will create a file named **Q1.sql** and create a file **Q2.sql for question 2**. So, if you do 10 questions, your folder must contain **only** 10 files Q1.sql, Q2.sql, Q3.sql, Q4.sql, Q5.sql, Q6.sql, Q7.sql, Q8.sql, Q9.sql and Q10.sql.

- Do not use any commands having the database name such as create database, alter database, use [database name], *etc*.

- Your response must contain only necessary commands for answering the question. Do not include any other command. For example, if you are required to create a trigger/procedure, then your response should contain only commands for creating the corresponding trigger/procedure; all commands for testing the created trigger/procedure are forbidden.

- Do not use GO command in your code

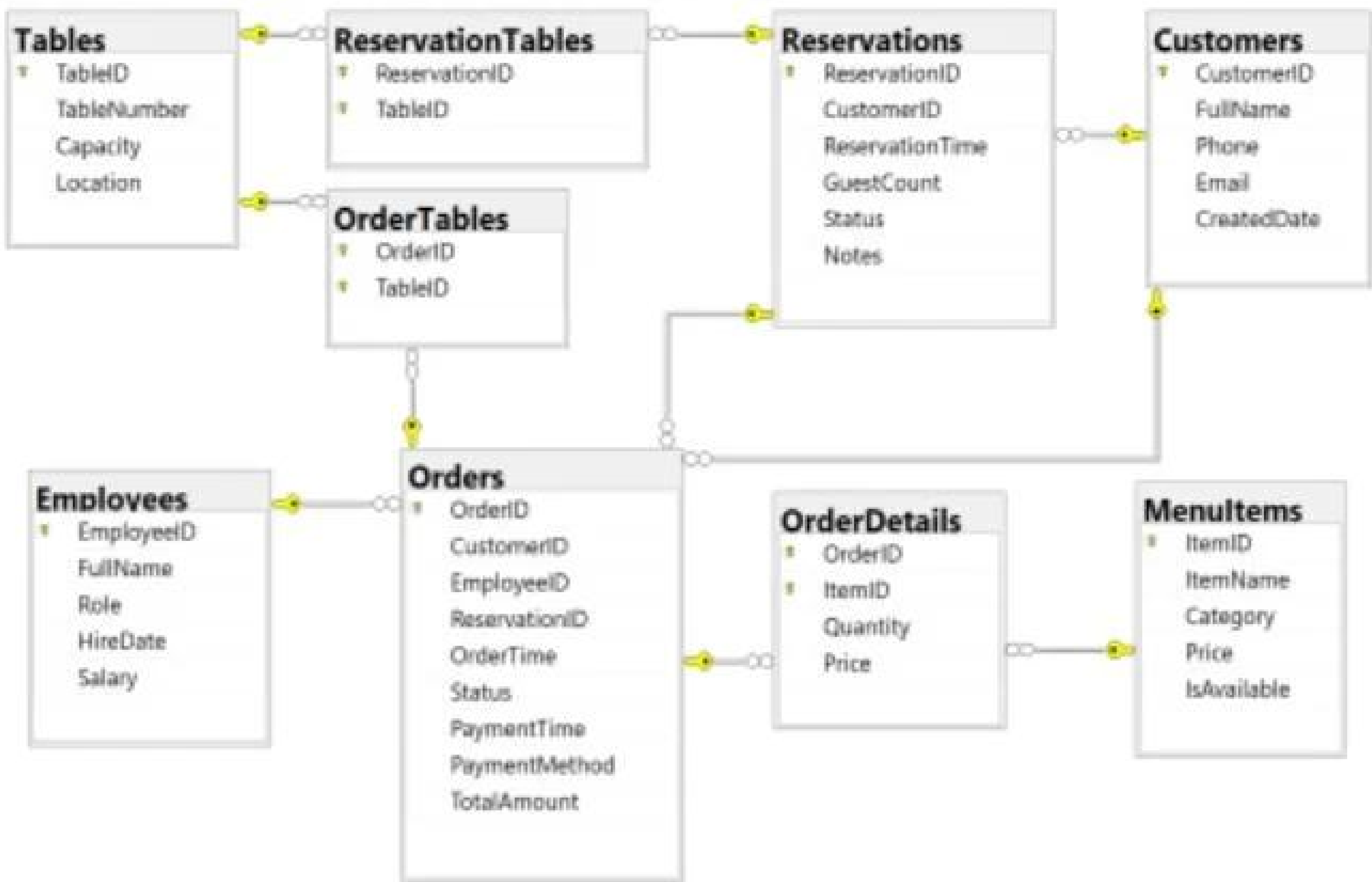
- On completion, import your work by browsing to the above folder.

- Note that:

+ You could use only SQL Server, SQL Server Management Studio, and paper or offline document in your computer.

+ If <u>any</u> of the previous requirements is not respected, your mark will be 0.

From the 2nd question, you should use the database provided in the .sql file which has the following database diagram. Please, run the provided script to create tables and insert data into your database.



Question 1:

Create one database and then write SQL statements to create, in this database, all tables derived from the ERD given in the following picture with appropriate attributes, primary keys and foreign keys.

NOTE that when creating the SQL commands as request, you MUST keep the name of tables, relationship, attributes and data type of attributes as SAME as given in the given ERD.

Attributes with underline are Primary Key of each entity.

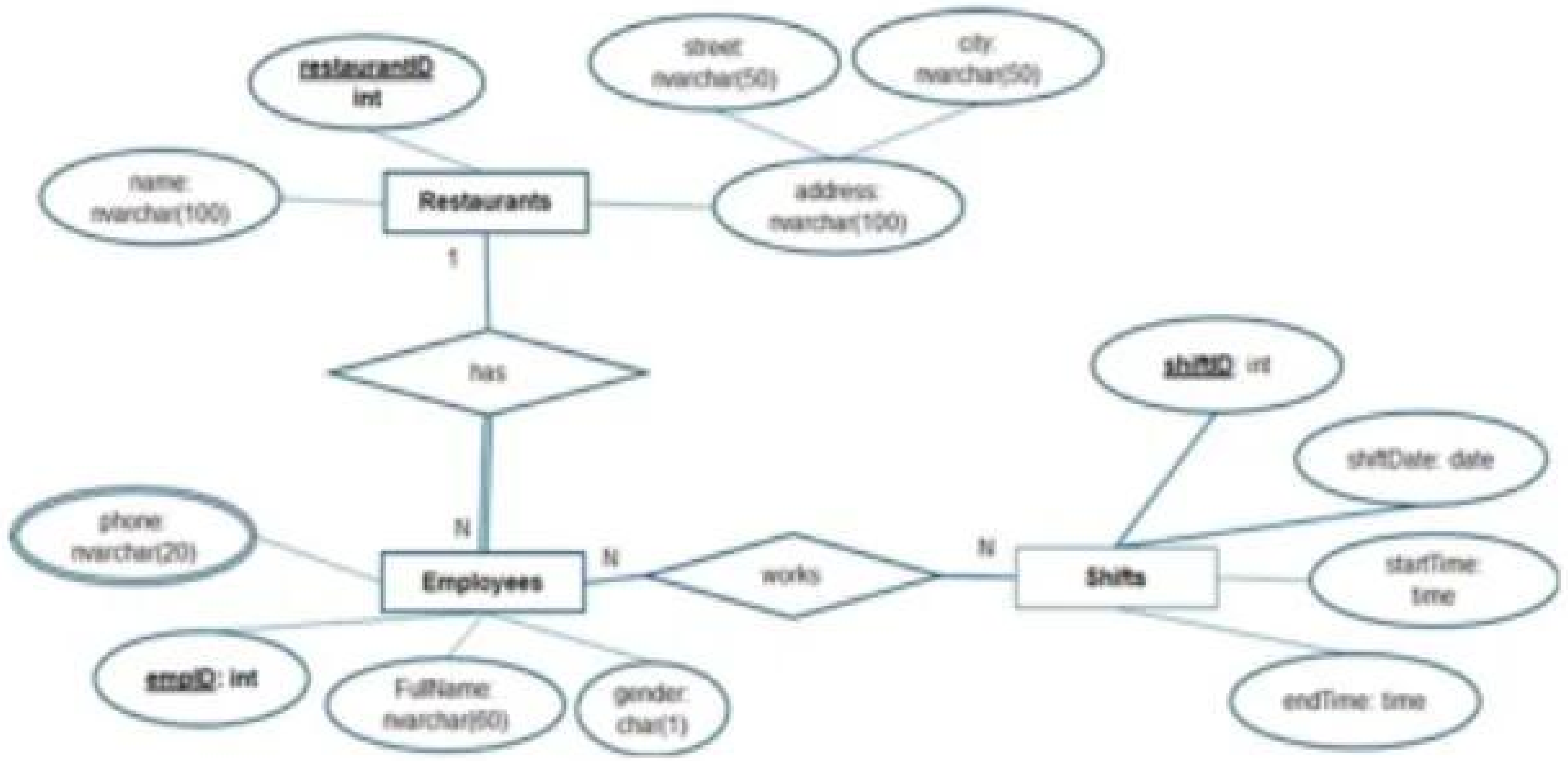
Attributes which reference to the primary key of another table must have the same name as the attributes in the primary key of the referencing table.

If a table must be created for representing a relationship, the name of the table must be the same as the name of the relationship.

When submitting the responses for this question, submit only SQL statements for creating tables with corresponding keys and foreign keys. Do not use "create database", "Alter database", "use database_name" or any statements using database's name in your submission.

You must use ER style conversions for subclasses in this diagram if exists.

The table represents the multi-values attribute must have the name which is the concatenation of the table name and the attribute name. For example, if table named Student has the multi-values attribute named Phone, the name of the corresponding table must be StudentPhone.



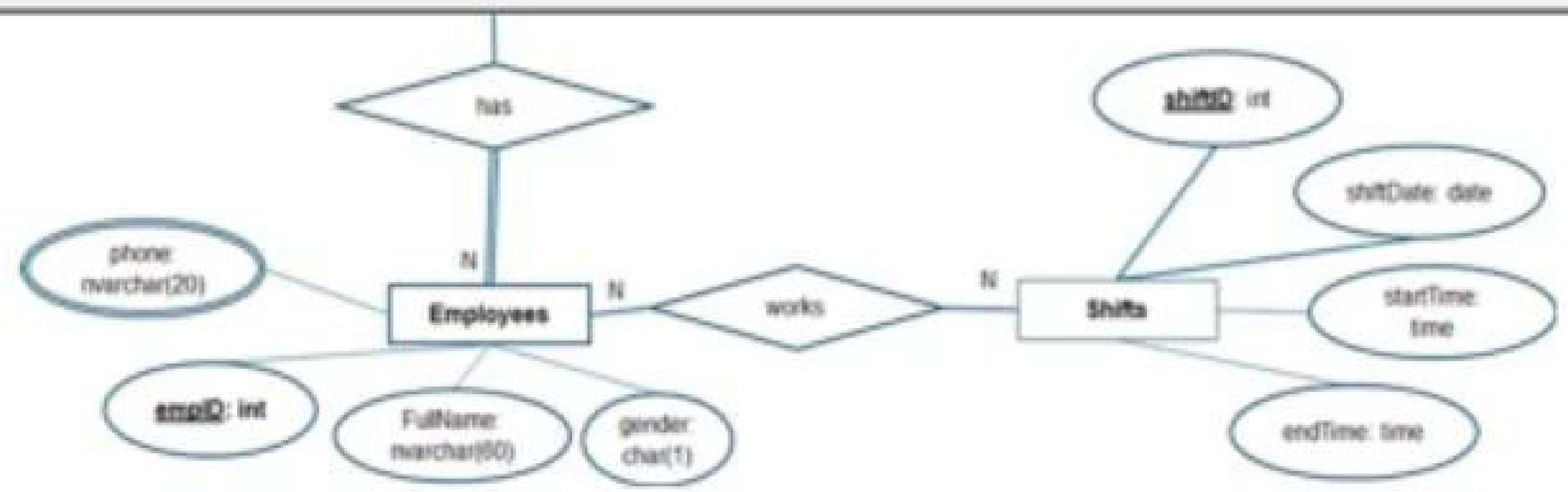
Picture 1.1

Question 2:

Zoom

\- + 70%

Close



Picture 1.1

Question 2:

Write an SQL query to display all employees who are chefs or managers and were hired in 2021, as shown in the following figure:

	EmployeeID	FullName	Role	HireDate	Salary
1	1	Nguyễn Văn An	Manager	2021-02-15	18000000.00
2	6	Nguyễn Thanh Tùng	Chef	2021-11-20	18500000.00
3	10	Phạm Quốc Việt	Chef	2021-08-09	19000000.00

Picture 2.1

Question 3:

Write an SQL query to list all menu items ordered by the customer named "Nguyễn Hoài Phương", displaying the CustomerID, FullName, ItemID, and ItemName. Ensure that duplicate records are not repeated in the result:

	CustomerID	FullName	ItemID	ItemName
1	36	Nguyễn Hoài Phương	3	Bún chả Hà Nội
2	36	Nguyễn Hoài Phương	5	Cơm rang dưa bò
3	36	Nguyễn Hoài Phương	11	Nộm đu đủ bò khô

Zoom

\- + 70%

Close

	EmployeeID	FullName	Role	HireDate	Salary
1	1	Nguyễn Văn An	Manager	2021-02-15	18000000.00
2	6	Nguyễn Thanh Tùng	Chef	2021-11-20	18500000.00
3	10	Phạm Quốc Việt	Chef	2021-08-09	19000000.00

Picture 2.1

Question 3:

Write an SQL query to list all menu items ordered by the customer named "Nguyễn Hoài Phương", displaying the CustomerID, FullName, ItemID, and ItemName. Ensure that duplicate records are not repeated in the result:

	CustomerID	FullName	ItemID	ItemName
1	36	Nguyễn Hoài Phương	3	Bún chả Hà Nội
2	36	Nguyễn Hoài Phương	5	Cơm rang dưa bò
3	36	Nguyễn Hoài Phương	11	Nộm đu đủ bò khô
4	36	Nguyễn Hoài Phương	14	Súp ngô gà
5	36	Nguyễn Hoài Phương	15	Súp nấm hải sản
6	36	Nguyễn Hoài Phương	16	Trà đá
7	36	Nguyễn Hoài Phương	17	Nước suối
8	36	Nguyễn Hoài Phương	19	Sinh tố bơ
9	36	Nguyễn Hoài Phương	23	Nước chanh
10	36	Nguyễn Hoài Phương	26	Kem dừa

Picture 3.1

Question 4:

Write an SQL query to display all main dishes (Category = 'Món chính'), along with the customers who ordered them in August 2025. Include menu items even if they were not ordered by any customer during that month. Show the following columns: Category, ItemID, ItemName, CustomerID, FullName. Sort the result by ItemName in ascending order, then by CustomerID in ascending order for rows of the same ItemName as shown in the following figure:

10	36	Nguyễn Hoài Phương	26	Kem dừa

Picture 3.1

Question 4:

Write an SQL query to display all main dishes (Category = 'Món chính'), along with the customers who ordered them in August 2025. Include menu items even if they were not ordered by any customer during that month. Show the following columns: Category, ItemID, ItemName, CustomerID, FullName. Sort the result by ItemName in ascending order, then by CustomerID in ascending order for rows of the same ItemName as shown in the following figure:

	Category	ItemID	ItemName	CustomerID	FullName
1	Món chính	4	Bún bò Huế	33	Đỗ Thị Hiền
2	Món chính	3	Bún chả Hà Nội	NULL	NULL
3	Món chính	5	Cơm rang dưa bò	44	Trần Hữu Nghĩa
4	Món chính	5	Cơm rang dưa bò	46	Phạm Văn Quang
5	Món chính	6	Cơm tấm sườn bì chả	NULL	NULL
6	Món chính	10	Gà rang muối	44	Trần Hữu Nghĩa
7	Món chính	9	Lẩu bò nhúng dấm	46	Phạm Văn Quang
8	Món chính	8	Lẩu thái hải sản	44	Trần Hữu Nghĩa
9	Món chính	7	Mì xào hải sản	NULL	NULL
10	Món chính	1	Phở bò tái	38	Lê Văn Hậu
11	Món chính	2	Phở gà	NULL	NULL

Picture 4.1

Question 5:

Write an SQL query to list all customers whose full name starts with "Nguyễn" or "Bùi".

Zoom

\- + 64%

Close

	Category	ItemID	ItemName	CustomerID	FullName
1	Món chính	4	Bún bò Huế	33	Đỗ Thị Hiền
2	Món chính	3	Bún chả Hà Nội	NULL	NULL
3	Món chính	5	Cơm rang dưa bò	44	Trần Hữu Nghĩa
4	Món chính	5	Cơm rang dưa bò	46	Phạm Văn Quang
5	Món chính	6	Cơm tấm sườn bì chả	NULL	NULL
6	Món chính	10	Gà rang muối	44	Trần Hữu Nghĩa
7	Món chính	9	Lẩu bò nhúng dấm	46	Phạm Văn Quang
8	Món chính	8	Lẩu thái hải sản	44	Trần Hữu Nghĩa
9	Món chính	7	Mì xào hải sản	NULL	NULL
10	Món chính	1	Phở bò tái	38	Lê Văn Hậu
11	Món chính	2	Phở gà	NULL	NULL

Picture 4.1

Question 5:

Write an SQL query to list all customers whose full name starts with "Nguyễn" or "Bùi", showing for each customer: CustomerID, FullName, the number of completed orders they made in the year 2025 (column NumberOfOrders), and the total amount spent on those orders (column TotalAmount). If a customer has no completed order in 2025, still display their name with NumberOfOrders = 0 and TotalAmount = 0.

	CustomerID	FullName	NumberOfOrders	TotalAmount
1	1	Nguyễn Thị Mai	2	1902000.00
2	7	Bùi Thị Ngọc	0	0.00
3	8	Nguyễn Hoàng Nam	1	573000.00
4	14	Bùi Văn Lâm	0	0.00
5	15	Nguyễn Thị Hồng	3	1576000.00
6	21	Bùi Thị Nhung	3	2156000.00
7	22	Nguyễn Văn Tín	1	900000.00
8	28	Bùi Ngọc Linh	1	570000.00
9	29	Nguyễn Văn Khánh	1	405000.00
10	35	Bùi Thị Trang	2	1155000.00
11	36	Nguyễn Hoài Phương	0	0.00
12	42	Bùi Thanh Bình	6	1947000.00
13	43	Nguyễn Lan Hương	0	0.00
14	49	Bùi Thị Thu Hà	3	1215000.00
15	50	Nguyễn Quốc Đạt	1	800000.00

Picture 5.1

Question 6:

Write an SQL query to list all menu items whose total sales amount in completed orders in 2023 is below the average total sales amount in 2023 of all menu items, including those with

Zoom

\- + 64%

Close

Picture 5.1

Question 6:

Write an SQL query to list all menu items whose total sales amount in completed orders in 2023 is below the average total sales amount in 2023 of all menu items, including those with

no sales in 2023. For each menu item, display: ItemID, ItemName, Category, TotalAmount (the total revenue generated from completed orders in 2023 (use 0 for items that were not sold in 2023). Ensure to calculate the average total sales across all items (including those with zero revenue). Sort the result by TotalAmount in ascending order, and then by ItemID in ascending order for items having the same TotalAmount.

	ItemID	ItemName	Category	TotalAmount
1	6	Cơm tấm sườn bì chả	Món chính	0.00
2	30	Bánh mì pate trứng	Món ăn nhẹ	40000.00
3	16	Trà đá	Thức uống	55000.00
4	18	Coca-Cola	Thức uống	60000.00
5	17	Nước suối	Thức uống	70000.00
6	7	Mì xào hải sản	Món chính	80000.00
7	27	Trà sữa trân châu	Thức uống	80000.00
8	21	Cafe đen đá	Thức uống	125000.00
9	28	Bánh chuối nướng	Tráng miệng	128000.00
10	22	Cafe sữa đá	Thức uống	140000.00
11	24	Chè thập cẩm	Tráng miệng	150000.00
12	29	Chè khúc bạch	Tráng miệng	165000.00
13	25	Bánh flan	Tráng miệng	200000.00
14	26	Kem dừa	Tráng miệng	210000.00
15	15	Súp nấm hải sản	Món khai vị	225000.00
16	19	Sinh tố bơ	Thức uống	245000.00
17	20	Sinh tố xoài	Thức uống	245000.00
18	23	Nước chanh	Thức uống	280000.00
19	12	Nem rán Hà Nội	Món khai vị	300000.00
20	14	Súp ngô gà	Món khai vị	320000.00
21	1	Phở bò tái	Món chính	325000.00
22	13	Chả giò hải sản	Món khai vị	420000.00

Picture 6.1

Question 7:

Write an SQL query to analyze customer spending trends between 2023 and 2024 based on completed orders. For each customer who placed completed orders in both 2023 and 2024, calculate and display:

- CustomerID

Zoom

\- + 54%

Close

Picture 6.1

Question 7:

Write an SQL query to analyze customer spending trends between 2023 and 2024 based on completed orders. For each customer who placed completed orders in both 2023 and 2024, calculate and display:

- CustomerID

- FullName

- AvgTotalOrderValue2023 - the average total order value of the customer's completed orders in 2023, rounded to two decimal places.

- AvgTotalOrderValue2024 - the average total order value of the customer's completed orders in 2024, rounded to two decimal places.

- PercentageChange - the percentage growth in the average total order value from 2023 to 2024. Il is calculated as (AvgTotalOrderValue2024 - AvgTotalOrderValue2023)/ AvgTotalOrderValue2023*100. Round the percentage change to two decimal places, and append ' %' at the end.

Order the result in ascending order of CustomerID as shown in the following figure.

	CustomerID	FullName	AvgTotalOrderValue2023	AvgTotalOrderValue2024	PercentageChange
1	2	Trần Văn Tâm	530000.00	340000.00	-35.85 %
2	12	Đỗ Đức Thịnh	475000.00	525000.00	10.53 %
3	13	Võ Thị Hoa	1820000.00	446000.00	-75.49 %
4	17	Lê Thị Diễm	1070500.00	235000.00	-78.05 %
5	43	Nguyễn Lan Hương	1705000.00	680000.00	-60.12 %
6	45	Lê Thị Tâm	347000.00	1102000.00	217.58 %
7	48	Võ Đức Tài	525000.00	255000.00	-51.43 %

Picture 7.1

Question 8:

Write a stored procedure named MonthlySalesSummary that generates a monthly sales report for a specified year. Your procedure must accept one input parameter: @year int - the year for which the report will be generated. The procedure must display all 12 months (from January to December), even if there are no orders in a given month. For each month, calculate and display:

- Month - in the format Month/Year

- NumberOfOrders - the total count of completed orders in that month

- TotalRevenue - the total revenue (sum of Quantity * Price) from all completed orders in that month

Picture 7.1

Question 8:

Write a stored procedure named MonthlySalesSummary that generates a monthly sales report for a specified year. Your procedure must accept one input parameter: @year int - the year for which the report will be generated. The procedure must display all 12 months (from January to December), even if there are no orders in a given month. For each month, calculate and display:

- Month - in the format Month/Year

- NumberOfOrders - the total count of completed orders in that month

- TotalRevenue - the total revenue (sum of Quantity * Price) from all completed orders in that month

For example, when we execute the MonthlySalesSummary procedure by using the following query, the results must be shown as in the following figure:

execute MonthlySalesSummary 2023

	Month	NumberOfOrders	TotalRevenue
1	1/2023	1	475000.00
2	2/2023	0	0.00
3	3/2023	4	3315000.00
4	4/2023	2	1390000.00
5	5/2023	1	575000.00
6	6/2023	4	2115000.00
7	7/2023	1	1105000.00
8	8/2023	1	1070000.00
[illegible]	[illegible]	[illegible]	[illegible]

Zoom

- + 56%

Close

```
execute MonthlySalesSummary 2023
```


	Month	NumberOfOrders	TotalRevenue
1	1/2023	1	475000.00
2	2/2023	0	0.00
3	3/2023	4	3315000.00
4	4/2023	2	1390000.00
5	5/2023	1	575000.00
6	6/2023	4	2115000.00
7	7/2023	1	1105000.00
8	8/2023	1	1070000.00
9	9/2023	1	347000.00
10	10/2023	2	2111000.00
11	11/2023	1	525000.00
12	12/2023	0	0.00

Picture 8.1

Question 9:

Create a trigger named trg_UpdateTotalAmount on the OrderDetails table to ensure the TotalAmount column in the Orders table always stays accurate.

Whenever an OrderDetails record is inserted, updated, or deleted, the trigger should automatically recalculate the total amount of the affected orders as the sum of (Quantity × Price) for all its items. If all OrderDetails of an order are deleted, the total amount of this order should be set to 0.

Zoom

- + 56%

Close

12	12/2023	0	0.00

Picture 8.1

Question 9:

Create a trigger named trg_UpdateTotalAmount on the OrderDetails table to ensure the TotalAmount column in the Orders table always stays accurate.

Whenever an OrderDetails record is inserted, updated, or deleted, the trigger should automatically recalculate the total amount of the affected orders as the sum of (Quantity × Price) for all its items. If all OrderDetails of an order are deleted, the total amount of this order should be set to 0.

For example, if you run the following queries on the table OrderDetails, the corresponding orders will be updated.

-- test case 1 -> TotalAmount of the orders with OrderID = 3 and OrderID = 5 will be updated

INSERT OrderDetails (OrderID, ItemID, Quantity, Price)

VALUES (3, 2, 3, 75000.00), (5,27,2,40000.00)

-- test case 2 -> TotalAmount of the order with OrderID = 6 will be updated

delete from OrderDetails where OrderID = 6 and ItemID = 1

Question 10:

Write SQL statements to create a new reservation for a customer. The reservation should be made for CustomerID = 7, with 5 guests, on November 20, 2025, at 19:15 PM. The ReservationID should be 210.

After inserting the reservation into the Reservations table, assign the reservation to a table in the ReservationTables table. The assigned table should be the one with the smallest TableID among all tables that have a capacity greater than or equal to 5.

Zoom

\- + 62%

Close